[12]R. D. Schuiling and B. W. Vink, Geochim. Cosmochim. Acta 31, 2399 (1967).
[13]K. V. Krishna Rao, S. V. Nagender Naidu and L. Iyengar, J. Amer. Ceram. Soc. 53, 124 (1970).
[14]W. F. Sullivan and S. S. Cole, J. Amer. Ceram. Soc. 42, 127 (1959).
[15]M. N. Danchevskaya *et al.*, Russ. J. Phys. Chem. 43, 444 (1969); I. A. Bondar *et al.*, Inorganic Materials (USSR) 7, 1183 (1971).

Continued from page 1759

[9]T. R. Sandin and P. H. Keesom, Phys. Rev. 177, 1370 (1969).
[10]J. S. Dugdale, J. A. Morrison and D. Patterson, Proc. Roy. Soc. (London) A224, 228 (1954).
[11]C. H. Shomate, J. Amer. Chem. Soc. 69, 218 (1947).
[12]H. J. McDonald and H. Seltz, J. Amer. Chem. Soc. 61, 2405 (1939).
[13]B. F. Naylor, J. Amer. Chem. Soc. 68, 1077 (1946).
[14]N. P. Slyusar, A. D. Krivorontenko, E. N. Fomichev, A. A. Kalashnik and V. P. Bondarenko, Teplofiz. Vys. Temp. 11, 213 (1973).
[15]J. Lietz, Hamburger Beitr. Angew. Mineral. u. Kristallphys. 1, 229 (1956).
[16]A. Navrotsky, J. C Jamieson and O. J. Kleppa, Science 158, 388 (1967).
[17]J. S. Arthur, J. Appl. Phys. 21, 732 (1950).
[18]M. S. Erofeeva, N. L. Lukinykh and S. M. Ariya, Russ. J. Phys. Chem. 35, 375 (1961).
[19]L. F. Nilson and O. Pettersson, Z. Phys. Chem. 1, 27 (1887).

Continued from page 1762

The bending frequency is estimated as 270 ± 80 cm^{-1} from the following information. A value of 272 cm^{-1} derived from the three-constant valence-force field[17] using ν_1, ν_3 and $(k\delta/l^2)/k_l = 0.040$ transferred from isoelectronic CaF_2.[18] The resulting value of k_{12}/k_l is similar to that of CaF_2. The inferior two-constant central-force field gives $\nu_2 = 325$ cm^{-1}. Bending modes have been observed[19] for electronic states of TaO_2 (285 and 281 cm^{-1}) and WO_2 (300 cm^{-1}). These molecules have stretching frequencies similar to TiO_2, but the excited states may be linear rather than bent as are the ground states. We estimate that $S°(298.15\ K)$ is uncertain by about 0.7 and 0.3 $cal \cdot K^{-1} \cdot mol^{-1}$ due to uncertainties in ν_2 and the geometry, respectively.

References
[1]P. J. Hampson and P. W. Gilles, J. Chem. Phys. 55, 3712 (1971).
[2]G. Balducci, G. DeMaria, M. Guido and V. Piacente, J. Chem. Phys. 56, 3422 (1972).
[3]J. Berkowitz, W. A. Chupka and M. G. Inghram, J. Phys. Chem. 61, 1569 (1957).
[4]P. G. Wahlbeck and P. W. Gilles, J. Chem. Phys. 46, 2465 (1967).
[5]W. O. Groves, M. Hoch and H. L. Johnston, J. Phys. Chem. 59, 127 (1955).
[6]G. A. Semenov, Inorganic Materials (USSR) 5, 54 (1969).
[7]JANAF Thermochemical Tables: TiO(g), TiO_2(rutile and g), $Ti_3O_5(\alpha, \beta)$, Ti_4O_7(cr), 12-31-73; O(g), 6-30-62.
[8]JANAF Thermochemical Tables, 2nd ed., NSRDS-NBS 37, (1971).
[9]K. Suzuki and K. Sambongi, Tetsu To Hagane 58, 1579 (1972).
[10]R. R. Merritt, B. G. Hyde, L. A. Bursill and D. K. Philp, Phil. Trans. Roy. Soc. London, Ser. A 274, 627 (1973).
[11]J. Drowart, P. Coppens and S. Smoes, J. Chem. Phys. 50, 1046 (1969).
[12]B. R. Conard and P. W. Gilles, presented at 162nd Natl. Meeting, Amer. Chem. Soc., Washington, D.C., (1971).
[13]E. G. Rauh and R. J. Ackermann, J. Chem. Phys. 60, 1396 (1974).
[14]N. S. McIntyre, K. R. Thompson and W. Weltner, J. Phys. Chem. 75, 3243 (1971); 69, 3488 (1965).
[15]R. L. DeKock and W. Weltner, J. Phys. Chem. 75, 514 (1971).
[16]M. Kaufman, J. Muenter and W. Klemperer, J. Chem. Phys. 47, 3365 (1967).
[17]G. Herzberg, "Infrared and Raman Spectra of Polyatomic Molecules," D. Van Nostrand Co., Inc., New York, pp. 186-187, (1962).
[18]V. Calder, D. E. Mann, K. S. Seshadri, M. Allavena and D. White, J. Chem. Phys. 51, 2093 (1969).
[19]W. Weltner and D. McLeod, J. Chem. Phys. 42, 882 (1965); J. Mol. Spectrosc. 17, 276 (1965).

Continued from page 1763

References
[1]J. Berkowitz, W. A. Chupka, and M. G. Inghram, J. Chem. Phys. 27, 87 (1957).
[2]S. A. Shchukarev, G. A. Semenov and K. E. Frantseva, Russ. J. Inorg. Chem. 4, 1217 (1959).
[3]K E. Frantseva and G. A Semenov, High Temp. 7, 52 (1969).

[4]M. Farber, O. M. Uly, and R. D. Srivastava, J. Chem. Phys. 56, 5312 (1972).
[5]JANAF Thermochemical Tables: V_2O_4(cr), V_2O_3(cr), VO(g), TaO(g), and TiO_2(g), 12-31-73.
[6]S. Killingbeck, Ph.D. Thesis, Univ. Kansas, 1964; Diss Abstr. 65, 1985 (1964).
[7]L. Brewer and G. M. Rosenblatt, Chem. Rev. 61, 257 (1961).
[8]M. Farber, R. D. Srivastava and O. M. Uy, J. Chem. Soc. Faraday I 68, 249 (1972).
[9]C. E. Moore, U. S. Nat. Bur. Stand. Circ. 467, Vol. I, (1949).

Continued from page 1779

References
[1]A. D. Mah and K. K. Kelley, U. S. Bur. Mines RI 5858, (1961).
[2]JANAF Thermochemical Tables: V_2O_5(cr), 6-30-73.
[3]F. V. Charlu and O. J. Kleppa, High Temp. Sci. 5, 260 (1973).
[4]H. Siemonsen and H. Ulich, Z. Electrochem. 46, 141 (1940).
[5]E. Vol'f and S. M. Ariya, Zhur. Obshchei. Khim. 29, 2470 (1959).
[6]W. G. Mixter, Amer J. Sci. 34, 141 (1912).
[7]O. Ruff and L. Friedrich, Z. Anorg. Chem. 89, 279 (1914).
[8]M. Kobayashi, Bull. Chem. Soc. Japan 8, 231 (1933).
[9]H. Muller, W. Biltz's Festschrift, Hannover, 1927; quoted by J. W. Mellor, "Comprehensive Treatise on Inorganic and Theoretical Chemistry," Vol. IX, Longman, Green and Co., London, (1929).
[10]R. A. Karasev, A. Yu. Polyakov and A. M. Samarin, Dokl. Akad. Nauk (SSSR) 85, 1313 (1952).
[11]H. M. Spencer and J. L. Justice, J. Amer. Chem. Soc. 56, 2306 (1934).
[12]C. T. Anderson, J. Amer. Chem. Soc. 58, 564 (1936).
[13]O. A. Cook, J. Amer. Chem. Soc. 69, 331 (1947).
[14]V. G. Khlyustov, A. S. Borukhovich, and V. A. Perelyaev, Soviet Physics Solid State 14, 1848 (1973).
[15]J. Jaffray and R. Lyand, Comp. rend 233, 133 (1951).
[16]M. Foex, S. Goldztaub, R. Wey, J Jeffray, and J. Wucher, J. recherches centre natl. recherche sci., Lab. Bellevue (Paris) 21, 237 (1952); Chem. Abstr. 47, 9080i (1953).
[17]J. Jaffray and A. Dumas, J. recherches centre natl. recherche sci., Lab. Bellevue (Paris) 5, 360 (1954); Chem. Abstr. 49, 6759d (1955).
[18]J. Stringer, J. Less Common Metals 8, 1 (1965).
[19]D. Adler, Solid State Phys. 21, 1 (1968).
[20]S. Minomura and H. Nagasaki, J. Phys. Soc. Japan 19, 131 (1964).
[21]I. G. Austin, Phil. Mag. 7, 961 (1962).
[22]J. Feinleib and W. Paul, Phys. Rev. 155, 841 (1967).
[23]M. Farber, O. M. Uy, and R. D. Srivastava, J. Chem. Phys. 56, 5312 (1972).

Continued from page 1790

[6]H. Muller, W Biltz's Festschrift, Hannover, 1927, quoted by J. W. Mellor, "Comprehensive Treatise on Inorganic and Theoretical Chemistry," Vol. IX, Longman, Green and Co., London, (1929).
[7]H. M. Spencer and J. L. Justice, J. Amer. Chem. Soc. 56, 2306 (1934).
[8]H Flood and O. J. Kleppa, J. Amer. Chem. Soc. 69, 998 (1947).
[9]C. T. Anderson, J. Amer. Chem. Soc. 58, 564 (1936).
[10]O. A. Cook, J. Amer. Chem. Soc. 69, 331 (1947).

Continued from page 1793

References
[1]A. Reisman and F. Holtzberg, High Temperature Oxides 2, 217 (1970).
[2]W. Muthmann, L. Weiss, and R. Riedelbanch, Ann. Chem. 55, 67, 82 (1907).
[3]J. E. Moose and S W. Parr, J. Amer. Chem. Soc. 46, 2656 (1924).
[4]A. Sieverts, A. Gotta and S. Halberstadt, Z. Anorg. Chem. 187, 159 (1930).
[5]G. Becker and W. A. Roth, Z. Phys. Chem. 167A, 16 (1933).
[6]B. Neumann, C. Kroger and H. Kunz, Z. anorg. allg. Chem. 218, 379 (1934).
[7]G. H. Humphrey, J Amer. Chem. Soc. 76, 978 (1954).
[8]E. J. Huber, Jr., E. L. Head, C. E. Holly, Jr., and A. L. Bowman, J. Phys. Chem. 67, 793 (1963).
[9]A N. Kornilov, V. Ya. Leonidov and S. M Skuratov, Zh. Fiz. Khim. 38, 2008 (1964).
[10]V. I. Smirnova and B F. Ormont, Russian J. Phys. Chem. 30, 1327 (1956).
[11]G. B. Barbi, J. Less Common Metals 22, 487 (1970).